



A review of anesthetic considerations and postoperative pain control after the Nuss procedure

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The Nuss procedure is named after Dr. Donald Nuss, who pioneered his technique for the repair of Pectus Excavatum in the late 1990's. In 1998, he published his 10-year experience with the minimally invasive repair.¹ Since then, hospitals across the world have adopted this method of repair for Pectus Excavatum, also known as Minimally Invasive Repair of Pectus Excavatum (MIRPE).

As the procedure has been modified, improved upon, and perfected, there have been advances in patient management from surgical management (see Notrica's review of modifications, page xx), to perioperative anesthetic care changes. This review will focus on the perioperative considerations for patients undergoing MIRPE from an anesthetic standpoint. Preoperative testing and interventions will be discussed, along with techniques utilized for intraoperative anesthetic management. Postoperative pain management and recovery will be addressed, which can be crucial in the care of this special patient population.

Pectus Excavatum is a common deformity of the chest wall seen in 1 in 300 to 400 births, with a male to female ratio of approximately 4–1.² Pectus Excavatum can result in multiple physiological abnormalities in the cardiopulmonary system, including diminished stroke volume and cardiac output, along with restrictive lung disease and ventilation-perfusion mismatching.³ Although patients may have concomitant medical conditions involving the connective tissues, including Marfan syndrome, the majority of patients with pectus excavatum are otherwise healthy.

Pre-pubertal children with minor deformities are typically asymptomatic and do not require surgical intervention. As children grow, however, the chest wall progressively becomes more rigid, and the pectus excavatum will frequently worsen. This worsening deformity will generally not affect patients at rest, but the cardiac and pulmonary effects become more pronounced with exercise. The resultant cardiopulmonary dysfunction should be focused on during the preoperative workup for this patient population.^{4,5} A complete review of the physiologic impact is included in the article by Obermeyer on pages xx).

Preoperative management

As part of the preoperative workup, multiple studies are performed to determine the level of cardiopulmonary compromise. After chest x-rays are obtained, patients will usually undergo a computed tomography (CT) or magnetic resonance (MR) scan to determine the degree of deformity. Patients experiencing any symptoms such as shortness of breath, chest pain, dyspnea on exertion, or fatigue with exercise should also undergo advanced cardiopulmonary testing. In addition to a baseline electrocardiogram (ECG), severely symptomatic patients should also undergo echocardiography. Echocardiography is often reserved for patients with new heart murmurs, arrhythmias, or irregular heart rhythms, but some centers include as part of their Marfan's workup. PFT's are generally in the low normal range, and will not change anesthetic or postoperative management for most patients.⁵

The CT scan of the chest can accurately measure the anterior-posterior distance and transverse dimensions of the chest at the level of greatest deformity. The anterior-posterior measurement is made at the point of maximum sternal inversion, between the sternum to the spine. The transverse measurement of the chest is made at this same point, and then a ratio between the AP and transverse measurements is calculated. This ratio is known as the Haller Index. A normal Haller index is 2.5, and a Haller Index above 3.2 is considered to be a moderate deformity. Haller indices above 3.5 are considered severe deformities. Both moderate and severe deformities can be considered for surgical correction.⁶ Correction index is becoming favored over the Haller index because it is less subject to normal variations in chest wall configuration. See Sujka and St Peter's review on pages xx.

Once the patient has undergone preoperative testing and is scheduled for surgery, there are a few important interventions to optimize care. Preoperative counseling about the operative approach, descriptions of the procedure by the surgical team, and educating patients and their families about what to expect regarding pain and speed of recovery of paramount importance for optimal recovery. Furthermore, a comprehensive description of pain management strategies and focused education on postoperative pain can help reduce patient and family concerns surrounding the procedure. Studies have demonstrated that these simple interventions with patients and families can lead to better surgical outcomes,

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decreased length of stay, and increased patient and family satisfaction.^{7–9}

In addition to preoperative counseling and education, there is data to support the use of preoperative gabapentin in the management of pain following major surgical procedures. A single dose of 900–1200 mg of gabapentin has been shown to be effective in reducing postoperative pain and opioid use.¹⁰ By reducing opioid use, patients who receive gabapentin also have fewer side effects from opioids, including urinary retention, constipation, nausea and vomiting, and respiratory depression.¹¹

Intraoperative management

Intraoperative management for patients with pectus excavatum varies based on patient comorbidities, and surgical techniques. Large bore intravenous access is needed to help resuscitate patients if any vascular injury occurs during the procedure. A double lumen endotracheal tube or bronchial blocker aids surgical exposure by complete deflation of the lung during thoracoscopy. While uncommon, serious adverse events reported during MIRPE, including dysrhythmias, arrhythmias, hypotension, ventricular puncture, hemothorax, pneumothorax, and mediastinal emphysema. For this reason, an arterial line is helpful to facilitate early identification and management of many of these pitfalls that can occur during the procedure. Nitrous oxide should not be used, as it can expand a surgically induced pneumothorax. Following completion of the procedure, a chest x-ray is obtained to ensure no significant pneumothorax is present. If the chest x-ray shows a clinically relevant pneumothorax, it allows for placement of a chest tube prior to leaving the operating theater.¹²

Positioning the patient is also important to help prevent injury to the brachial plexus. The combination of vertical rolls to push the chest forward with abduction of both arms on armboards increases the risk of a brachial plexus injury. Alternative positioning techniques have been described, including adduction of both arms with vertical chest rolls, and the use of an arthroscopy sling to allow for arms to be extended above the head at an angle of less than 90°. Regardless of the technique chosen, it is important to position the patient in a way that minimizes the risk of injury to the brachial plexus.¹³

Postoperative management

MIRPE can result in significant postoperative pain, which becomes difficult to treat if not carefully managed. In addition to preventing medical and surgical complications, appropriate management of postoperative pain is necessary to help patients regain function, increase patient satisfaction, and hasten recovery. If postoperative pain is poorly controlled, it can delay recovery from surgery, may lead to familial and patient anxiety, delays participation in physical therapy, and increases hospital length of stay.¹⁴

Management of postoperative pain has been studied extensively, however, there is still no consensus on a single best approach in treating patients undergoing MIRPE. Whether utilizing regional anesthetics, surgical catheter placement, cryoablation, or multimodal approaches, all methods suggest the utilization of more than just opioid analgesia.¹⁵ Each approach will be discussed, along with the advantages and disadvantages of each.

Discussion of anesthetic options

For patients who receive a regional anesthetic in conjunction to a general anesthetic for MIRPE, there are two main options: thoracic epidural placement, or paravertebral catheter placement. Both types of catheters can be placed in cooperative patients while awake, however, many pediatric institutions that utilize regional

techniques will perform these blocks after induction of anesthesia.^{16–17}

Thoracic epidurals

Thoracic epidural analgesia can provide excellent analgesia following MIRPE, however, there are several issues that must be addressed. An anesthesiologist or pain management team familiar with safe regional techniques and protocols must carefully manage analgesia provided by a thoracic epidural. Epidural infusions typically include the use of a local anesthetic, along with an adjuvant, such as an opioid or clonidine. Epidural analgesia may not provide complete relief from pain, and, unfortunately, may occasionally result in unilateral instead of bilateral analgesia. For this reason, epidural analgesia frequently requires supplemental opioids, either through a patient controlled analgesia (PCA) pump, or intermittent opioids given intravenously. Epidurals that do not provide complete pain relief may also result in the need for manipulation of the epidural catheter postoperatively, which can be challenging in the immediate postoperative period. As with other types of epidural analgesia, a thoracic epidural infusion may necessitate the placement of a foley catheter, and delay ambulation while in place due to lower extremity weakness and instability.^{18–19}

In most studies, patients receiving a thoracic epidural as their primary pain management strategy had difficulty when transitioning from epidural analgesia to opioid analgesia.^{1,2} This difficult transition can result in increased pain scores several days after surgery, and will frequently delay discharge. Unlike lumbar epidural analgesia, thoracic epidural analgesia carries a higher risk of neurological complications due to the location of the spinal cord in relation to the insertion point of the catheter.³ Due to the location of the sympathetic chain in the thoracic spine, epidural infusions in this area can also result in hypotension and hemodynamic compromise. Finally, there is a discrete failure rate in placement of thoracic epidurals, which can range from 5 to 30%, depending on provider experience.^{20–21} In the randomized controlled trial by St Peter, et al, they found no benefit for thoracic epidural over PCA, nor did the randomized controlled trial by Jaroszewski et al.^{4,5} Retrospective studies have also come to the same conclusion.^{1,2}

Paravertebral blocks

With the widening use of ultrasound guided regional anesthesia, paravertebral blocks and paravertebral catheter placement have become viable alternatives to thoracic epidural catheter placement. Paravertebral blocks are placed lateral to the transverse processes of the spinal cord and allow for spread of local anesthetic to multiple intercostal nerves, with potential spread to the epidural space as well. Although ultrasound guided placement reduces

¹ Singhal NR, Jones J, Semenova J, et al. Multimodal anesthesia with the addition of methadone is superior to epidural analgesia: A retrospective comparison of intraoperative anesthetic techniques and pain management for 124 pediatric patients undergoing the Nuss procedure. *J Pediatr Surg* 2016;51(4):612–616.

² Man JY, Gurnaney HG, Dubow SR, et al. A retrospective comparison of thoracic epidural infusion and multimodal analgesia protocol for pain management following the minimally invasive repair of pectus excavatum. *Paediatr Anaesth* 2017;27(12):1227–1234

³ Kelly RE, Goretsky MJ, Obermeyer R, et al. Twenty-one years of experience with minimally invasive repair of pectus excavatum by the Nuss procedure in 1215 patients. *Ann Surg* 2010;252(6):1072–1081.

⁴ St Peter SD, Weesner KA, Weissend EE, et al. Epidural vs patient-controlled analgesia for postoperative pain after pectus excavatum repair: a prospective, randomized trial. *J Pediatr Surg* 2012;47(1):148–153.

⁵ Jaroszewski DE, Temkit M, Ewais MM, et al. Randomized trial of epidural vs. subcutaneous catheters for managing pain after modified Nuss in adults. *J Thoracic Dis* 2016;8(8):2102–2110.

risks during catheter placement, there is a chance for a pneumothorax during catheter placement, since the paravertebral space is just superficial to the pleural space. This risk is eliminated once the thoracoscopy for the MIRPE has begun, since the procedure involves the surgeon creating a controlled pneumothorax bilaterally. Spread to the epidural space is uncommon, and as a result, there is a much lower rate of hypotension and hemodynamic compromise with paravertebral catheter placement when compared to thoracic epidural catheters.²² This can result in effective analgesia for MIRPE, without some of the drawbacks seen with thoracic epidural placement. Similar to thoracic epidural catheter placement, however, the placement of paravertebral catheters is an advanced regional anesthetic technique and may not be an option for many hospital systems. Furthermore, two catheters would have to be placed to allow for bilateral analgesia.²³

Elastomeric pain infusion catheters

Two main types of surgical interventions have been described to help with postoperative pain control: intraoperative catheter placement, and cryoablation. Intraoperative catheter placement is most commonly done in a subpleural fashion to help anesthetize intercostal nerves, or subcutaneously to help with incisional pain and potentially anesthetize intercostal nerves as well. The subpleural catheter is tunneled through a tiny incision and placed under the pleura under direct visualization with the thoracoscope. This can lead to increased surgical time, as tunneling the catheter under the pleura is a delicate process, which could lead to a pneumothorax if done incorrectly. After the catheters are tunneled, they are attached to reservoirs containing local anesthetics, which infuse over the course of several days. While intraoperative catheter placement directly into the surgical site may also reduce postoperative pain, this position has been associated with a higher rate of surgical site infection and should be used with caution. Additionally, the local anesthetic must be carefully calculated and ordered to prevent overdose of the local anesthetic and local anesthetic toxicity.²⁴

Cryoablation

Cryoablation of intercostal nerves has been a novel approach to addressing postoperative pain, and has been utilized more frequently over the past few years. There have been several approaches described, but it is most commonly utilized to temporarily damage the intercostal nerves at multiple levels on both sides of the chest wall. Based on the extent of the surgery and the number of bars placed, the provider can choose to perform ablation at the level of the incision, and either one or two levels above and below the initial ablation site. This results in either three or five levels of cryoablation on each side of the chest. Cryoablation has been shown to provide equivalent analgesia to thoracic epidurals, and does not have the drawbacks associated with catheter placement. Additionally, the procedure can be performed through the same incisions used for the surgical procedure. Cryoablation does result in an increased cost to the patient, increased surgical time, and the need for the surgical providers to learn a technique still being explored. Although long term outcomes for cryoablation are unavailable, there is the possibility of persistent neuropathic pain. This may potentially be mitigated by the use of perioperative gabapentin. In most studies, the intercostal nerves regenerate within three months following the ablation procedure.²⁵ Cryoablation has been shown to reduce perioperative opioid use, reduce length of stay, and reduce perioperative pain after a single intraoperative treatment.²⁵

Multimodal anesthesia

In centers that do not have access to the advanced interventional techniques described above, there are alternatives to management of postoperative pain in patients undergoing MIRPE. Multimodal analgesia is an important and effective method of treating even the most severe postoperative pain. In treating patients with a multimodal analgesia, there are two types of approaches, preventive, and preemptive. Both methods are aimed at reducing central sensitization from surgical interventions, and both try to reduce the overall need for analgesics postoperatively. Surgical incisions induce central sensitization, and this is composed of the incisional phase and the inflammatory phase. Preemptive analgesia refers to medications and therapies administered prior to surgical incision, and it remains an area of focus that needs further investigation. It aims to reduce the inflammatory mediators released in response to surgical trauma however its efficacy is unclear. Mixed data exist and preemptive analgesia has not been shown to be more effective than preventive analgesia.²⁶

Preventive analgesia, on the other hand, is a multimodal approach geared towards reducing overall use of analgesics, and has been shown to be the most effective method of treating postoperative pain. Preventive analgesia is geared at reducing both responses to the incisional phase, as well as the inflammatory phase in response to surgery. By starting interventions in the preoperative and intraoperative portions of the surgery, and continuing a multimodal approach postoperatively, preventive analgesia can reduce overall use of opioids and need for supplemental pain medications.^{26–27}

Several medications can be utilized to help with preventive analgesia, many of which may not be thought of as first line therapies for pain management. According to the World Health Organization (WHO) Stepladder for pain, acute pain management should occur in a stepwise fashion. For mild pain, starting therapies should include a combination of non-steroidal anti-inflammatory drugs (NSAID's), and acetaminophen. For moderate pain, it is recommended to add oral opioids to the baseline medications given. For severe pain, intravenous opioids are recommended.²⁸

Although the WHO stepladder helped bring into focus the need for medications other than opioids in dealing with pain management, many centers have gone even further in multimodal approaches. Several societies, including the American Pain Society, American Society of Anesthesiologists, and the American Society of Regional Anesthesia and Pain Medicine have discussed the importance of pain management in the perioperative period. By aggressively managing pain, patients can have better outcomes, a reduction in complications from surgery, a quicker return to normal function, and better overall satisfaction. Scheduled acetaminophen and ketorolac can reduce the amount of opioid needed postoperatively. These medications can also be safely administered during the period of most pain, generally seen in the first three postoperative days.^{29–30}

There are several medications that can be utilized aside from acetaminophen, NSAIDs, and opioids. In patients undergoing MIRPE, the use of these adjuvant and supplement non-opioid medications can result in significant reduction in opioid use postoperatively.³¹ Commonly used NMDA receptor antagonists include ketamine and dextromethorphan. Methadone is an opioid medication that also exhibits NMDA receptor antagonism, and can be useful in the treatment of surgical pain. The use of ketamine as a multimodal analgesic has been well established, and single doses used intraoperatively have been shown to reduce opioid use and pain scores for up to 48 h.³² The combination of methadone and ketamine is even more effective, reducing overall opioid use in the five days after surgery.³³

Methadone is an opioid that can have a half life that is quite variable between patients, from 24–72 h, but this is seen in patients receiving scheduled methadone over the course of several days to weeks. A bolus dose given in the OR, however, will not result in the stacking phenomenon that can be seen in patients on long term methadone therapy. Patients undergoing MIRPE will have an electrocardiogram performed as part of their preoperative workup. Assuming a normal QTc on patient EKG, it is safe to administer a single dose of 0.1–0.3 mg/kg of methadone to assist with postoperative pain control.³⁴

Intraoperative dexamethasone has also been shown to reduce opioid use post-operatively. A single dose of 0.1–0.2 mg/kg has been shown to reduce pain scores for over 24 h after administration. This low dose also reduces untoward side effects such as hyperglycemia and the potential for wound infections. As a potent anti-inflammatory, its effects can last well beyond its clinical half life, especially since it acts on the inflammatory phase of central sensitization seen after surgical interventions. In addition, dexamethasone also is an excellent anti-emetic, which can be helpful for patients receiving opioid therapies for pain control.³⁵

In addition to methadone, ketamine, and dexamethasone as adjunctive therapies during MIRPE, the use of clonidine has been shown to help reduce opioid use postoperatively. Clonidine, an α -2 receptor agonist, helps to reduce circulating norepinephrine by acting on centrally located α receptors. In addition to having effects on blood pressure, clonidine can reduce anxiety, and augment the effects of other analgesic medications. The use of a clonidine patch postoperatively can help reduce the risk of hypotension seen with bolus dosing, and can be used to help with opioid sparing pain management strategies. Clonidine also offers an advantage over dexmedetomidine, due to the reduced risk of bradyarrhythmias that can occur with dexmedetomidine infusions. Many institutions require continuous cardiac monitoring for patients with dexmedetomidine infusions, which is unnecessary with a clonidine patch.³⁶

Diazepam is another adjunctive medication that can be utilized post-operatively to help with pain management. As a benzodiazepine, diazepam works on GABA receptors, and can help with the treatment of muscle spasms, anxiety, and nausea. Additionally, the effects on the GABA receptor can help to augment the effects of other non-opioid and opioid pain medications. In patients undergoing MIRPE, the muscles in the chest wall can be significantly and painfully stretched, and this pain is not always addressed by the other medications discussed. While patients can receive this intravenously as needed in the first 24 h following surgery, it should be converted to an oral form as soon as possible.³⁷

When utilizing a multimodal approach without regional anesthesia following a MIRPE, it is important to ensure that adequate opioids are available for pain control. To help ensure a smooth transition from the operating room to the recovery area, a PCA pump is recommended to allow for increased patient comfort and patient control. In addition, understanding patient use in the first 24 h can help guide management for the remainder of the hospital stay. Patients can be converted from intravenous opioids via the PCA pump to an oral regimen on the first postoperative day. This can be achieved through the use of long acting oral opioids, such as oxycodone SR or morphine SR, along with immediate release opioids such as oxycodone or hydrocodone.³⁸

As with any opioid therapy, the side effects of urinary retention, respiratory depression, and nausea and vomiting must be carefully balanced with pain management. Particularly important is the management of nausea and vomiting, which needs to be controlled in order for patients to tolerate oral therapies. In addition to ondansetron, a 5HT-3 receptor antagonist, other anti-emetic therapies may be considered for patients having significant issues with nausea and/or vomiting. Antihistamines such as diphenhydramine can

also be considered as adjunctive therapies. Promethazine, which is an H1 antagonist and anticholinergic can also be considered for refractory nausea. Finally, a scopolamine patch can be applied transdermally for 72 h in the case of ineffective control of nausea from intermittent therapies.³⁹

Finally, early mobilization and physical therapy following surgical intervention is important. With appropriate pain control following MIRPE, patients can safely move from their bed to a sitting position on the first postoperative day. With aggressive physical therapy and assistance, most patients will be ambulatory by the second postoperative day. Early mobilization can help reduce postoperative complications seen with prolonged bedrest, and can help patients and families become functional more quickly.⁴⁰

Conclusions

MIRPE is a procedure performed nationwide with many different analgesic approaches taken from both a surgical and anesthetic perspective. Much debate has occurred on how to best take care of this patient population postoperatively. Although there is no single best method, it is clear that effective management begins in the preoperative period. Multimodal analgesia can be reproduced without the need for advanced surgical or anesthetic techniques, while at the same time accomplishing excellent analgesic goals. Newer adjuncts such as cryoanalgesia for long-term pain control are also being reported.

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